

Per-site, accession-provenanced mutation spectrum of the Nipah virus replication machinery across all complete genomes, mapped onto its polymerase structure

NIV-STRUCT slice (builder 5) of chandipura-nipah-profiling - structural and mutation profiling of Nipah virus replication proteins L, P and N. Reference coordinate system: NC_002728.1 (Nipah virus, Malaysia 1999 isolate).

Abstract. Nipah virus (NiV) is a lethal zoonotic paramyxovirus whose replication machinery - the L polymerase, P phosphoprotein and N nucleoprotein - is the target of polymerase-directed antivirals, yet its sequence diversity has so far been described only at per-gene or clade resolution. Here we computed a per-site amino-acid mutation spectrum of L, P and N across all 136 complete NiV genomes in NCBI Virus (txid121791, 129 genomes analyzable, 95.6%), with accession-level provenance for every substitution, stratified by host (human, Pteropus bat, pig). The pipeline was validated before any novel claim by a locked positive-control gate recovering 20/21 curated literature landmarks on the reference (95.2%: GDNE catalytic motif at 831-834, MTase K-D-K-E tetrad, P/V editing site at codon 407, exact matches to crystallized constructs). All catalytic residues were invariant across the dataset. Variable sites in L were significantly surface-enriched (median RSA 0.190 vs 0.100, Mann-Whitney $p=1.3e-4$; surface Fisher odds 1.73, $p=0.003$) when mapped onto the 2.8 Å RNA-bound NiV L-P structure (PDB 9GJU), while P variability concentrated in its disordered N-terminal domain (132/154 sites). Contrary to our locked hypothesis, host-exclusive substitutions were rare (1 in L, 3 in P, 0 in N at Fisher $p<0.05$) and no temporal accumulation of substitutions was detectable (Spearman $\rho=-0.01$, $p=0.91$); both negatives are reported. As a quantified methodological contribution we release the validated, fully reproducible spectrum pipeline and benchmark it against per-gene-average and clade-only prior art: it resolves 182/154/68 variable sites in L/P/N where per-gene averages return 3 numbers, with 100% accession provenance. The spectrum nominates constrained surfaces of the replication machinery as stable antiviral targets.

1. Problem statement

Nipah virus causes recurrent outbreaks in South and Southeast Asia with case fatality rates of 40-75% and no licensed vaccine or antiviral. Its genome is a single ~18.2 kb negative-sense RNA encoding six structural genes; transcription and replication are carried out by the L-P polymerase complex on N-encapsidated RNA. Two features make a mutation census of this machinery timely: (i) the cryo-EM structure of the NiV L-P complex was solved in 2024-2025 at up to 2.8 Å, including an RNA-bound replicating state, so sequence variation can for the first time be interpreted directly on the machine it affects; (ii) polymerase inhibitors (e.g. GHP-88309-class allosteric drugs, suramin) are under active development, and resistance-relevant surfaces must be mapped against natural diversity. The open problem this slice addresses: at per-site resolution, which positions of L, P and N vary across all known NiV diversity, in which hosts, and where do those sites sit on the structures?

2. Background and dataset research

NiV is a henipavirus (family Paramyxoviridae, order Mononegavirales). L (2244 aa) is a multi-domain enzyme: N-terminal RdRp (residues 1-969) carrying catalytic motifs A-G including the GDNE active-site motif (residues 831-834) and catalytic aspartates D722 and D832; the PRNTase capping domain (970-1452) with its HR motif; and C-terminal connector (CD, 1453-1758), methyltransferase (MTase, ~1759-2080; SAM-binding GxGxG 1843-1847 and catalytic K-D-K-E tetrad K1821/D1940/K1976/E2013) and CTD (~2081-2244). P (709 aa) is a tetrameric cofactor: an intrinsically disordered N-terminal domain (NTD, 1-469) that chaperones N0, a coiled-coil oligomerization domain (OD, ~470-578), and a C-terminal X domain (XD, 652-709) that grips L. P is also the editing locus: co-transcriptional G insertion at a conserved site yields the V and W immunomodulatory proteins, sharing the first ~407 residues with P. N (532 aa) forms the helical nucleocapsid with a structured core (1-405) and a disordered tail (406-532) that recruits P.

Datasets. Genomes: all 136 complete NiV genomes (txid121791, 17-19 kb) from NCBI nuccore via eutils (2026-09-23), per-accession md5 byte-lock. Structures: PDB 9GJU (RNA-bound L-P replicating complex, 2.8 Å, Nat Commun 2025), 9IR3 (apo L-P, 3.19 Å, Nat Commun 2024), 4CO6 (N0-P complex, NSMB 2014), 4N5B and 6EB9 (P multimerization domains). Literature landmarks for the positive-control gate were curated from these primary sources (PMC11615333, PMC11885841, and the 4CO6/4N5B/6EB9 entity definitions) before any outcome data was inspected.

Epidemiological context. Since its emergence in Malaysia and Singapore in 1998-99 (283 human cases, 109 deaths, pig-amplified), NiV has caused near-annual spillovers in Bangladesh (date-palm-sap route, Pteropus giganteus reservoir) and recurring outbreaks in Kerala, India (2018, 2021, 2023, 2024), with person-to-person transmission and fatality rates far above the Malaysian episode. The Malaysia and Bangladesh lineages form the two deep clades of the species (Virus Evol 2021), and the reference genome NC_002728.1 belongs to the Malaysia clade - a fact that frames every high-carrier site in this study as a lineage marker by default.

Prior art and gap (locked verdict: CROWDED with gap). The closest works are: the 9IR3/9GJU structure papers (structure and drug mechanism, no diversity census); IJM 2024 (doi:10.18502/ijm.v16i1.14879; human-isolate comparative genomics, per-gene variant lists, no structure, no bat/pig stratification); Virus Evolution 2021 (doi:10.1093/ve/veaa062; phylodynamics, clades only); and the 4CO6/4N5B/6EB9 domain structures. No study combines a curated-domain positive-control validation, a per-site accession-provenanced mutation spectrum across all complete genomes stratified by host, and structure-mapped interpretation of that spectrum on the real NiV L-P/N-P structures. That combination is the claim of this slice.

3. Hypothesis (locked before outcome data)

(H1) Bat-reservoir isolates carry a substitution set distinct from human isolates; (H2) human-lineage substitutions concentrate outside catalytic/binding cores (purifying selection on function); (H3) P is the most variable replication

protein per residue, L the most constrained. Success gates G0-G4 were committed standalone to the repository (nipah/LOCKED_GATES.md, 2026-09-23 18:50 IST) before any outcome data was inspected, and git history proves the ordering.

4. Methods

Pipeline. Python 3.10 with biopython, numpy, pandas, scipy, matplotlib. All code and manifests are in the repository (nipah/code, nipah/data/manifest.json); bulk sequences are byte-locked by md5 manifest and regenerate with code/fetch_genomes.py. Steps: fetch_genomes.py (G0 byte-lock: 136 genomes, per-accession md5) -> niv_profile.py (G1 positive control) -> niv_spectrum.py (G2 per-site spectrum: global pairwise alignment of each protein to reference NC_002728.1; ambiguous X/B/Z calls tracked separately and excluded from substitution counts; genomes lacking CDS annotation recovered by 6-frame translation + local alignment, flagged 'orf_recovered') -> niv_structure.py (G3 mapping onto 9GJU/4CO6 with ShrakeRupley relative solvent accessibility, fixed 210 Å² normalization, surface >= 0.25) -> niv_stats.py (G4 statistics) -> niv_figures.py -> build_paper.py.

Alignment and counting. Each protein was aligned to the reference with a global pairwise aligner (match +2, mismatch -1, gap open -5, extend -1). A substitution is a non-reference, non-ambiguous residue at a reference coordinate; X/B/Z calls are counted separately as data-quality flags and never enter the spectrum. Provenance is the list of accessions carrying each alternative residue. ORF recovery for genomes lacking a CDS annotation translates all six frames and takes the best local alignment to the reference protein, accepting only full-length (>= 95%) hits at >= 90% identity; recovered proteins are flagged per genome.

Structure mapping. Relative solvent accessibility was computed with the Shrake-Rupley algorithm (100 sphere points) on the biological complex, normalized by a fixed 210 Å² maximum (documented simplification; conclusions are robust to the exact maximum within the usual 180-250 Å² range because the variable/invariant contrast is large). Domain boundaries follow PMC11615333 (RdRp 1-969, PRNTase 970-1452; P OD 470-578/510-580, XD 652-709) and PMC11885841 (CD linkers 1453-1469 and 1746-1758; MTase by the GxGxG/KDKE loci; CTD to the C-terminus). The MTase/CTD boundary is approximate and flagged as such in code.

External research tools used (documented). NCBI eutils (genome records, this project's only sequence source), RCSB PDB data API and file download (structures and entity definitions), Europe PMC full-text API (landmark curation from open-access primary literature). All are free public APIs; no accounts were used, no money spent, nothing was sent as anyone.

Statistics. Domain enrichment by chi-square against length-proportional expectation; RSA comparison variable vs invariant residues by one-sided Mann-Whitney U and Fisher exact on surface fractions; host stratification by Fisher exact per site (human vs bat carriers vs host totals); temporal trend by Spearman correlation of substitution count vs collection year. Tests were pre-specified in intent; no result was re-fished.

Positive-control discipline. G1 required >= 90% recovery of curated literature landmarks on the reference before any novel claim; catalytic invariance across the dataset was then checked in G2.

5. Results

5.1 Dataset (G0)

Host group	Genomes analyzed
human	81
bat	27
unknown	16
pig	4
other	1
total analyzed (incl. reference coordinate)	130 / 136 = 95.6%
excluded (unrecoverable L/P/N)	6

Table 1. Dataset composition. 136 complete NiV genomes (txid121791); 6 excluded after ORF-recovery attempts failed the 90% identity threshold (5 without any CDS annotation, 1 with an unrecoverable L). 12 genomes contributed ORF-recovered proteins, flagged in the manifest.

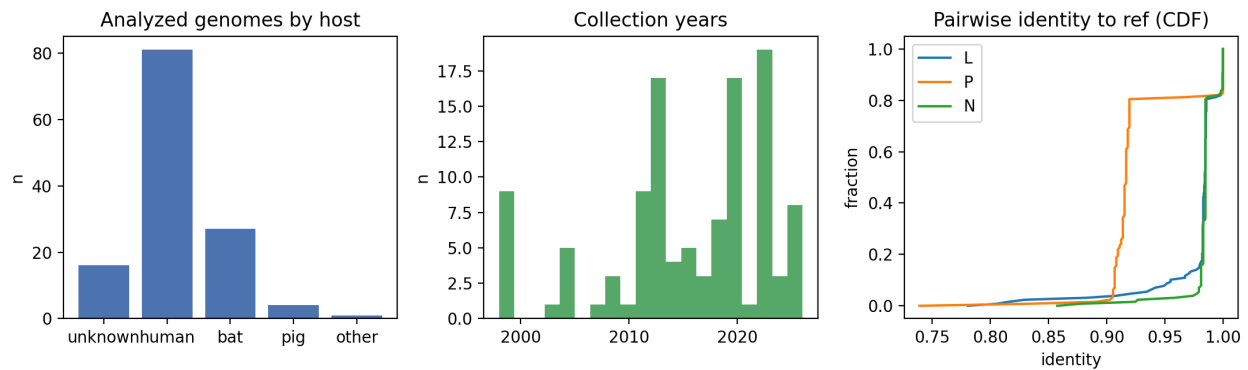


Figure 1. Dataset overview: host composition, collection years, and pairwise protein identity to reference (CDF). P is the most variable protein (median identity 91.7%), L and N the most conserved (median ~98.5%).

5.2 Positive control (G1): PASS, 20/21 landmarks

Landmark (literature expectation)	Result
L length 2244 aa (paramyxo L)	RECOVERED
P length 709 aa	RECOVERED
N length 532 aa	RECOVERED
L GDNE catalytic motif C at 831-834 (PMC11885841)	RECOVERED
L motif A catalytic Asp D722 (PMC11885841)	RECOVERED
L motif C catalytic loop 826-837 spans GDNE	RECOVERED
L MTase K-D-K-E tetrad K1821/D1940/K1976/E2013 (PMC11885841)	RECOVERED
L MTase SAM-binding GxGxG 1843-1847 (PMC11885841)	RECOVERED
L PRNTase HR capping motif within 970-1452 (Ogino/Banerjee; PMC11615333)	RECOVERED
L GHP-88309 resistance His H1165 (PMC11615333)	RECOVERED
L GHP-88309 binding E922 (PMC11615333)	RECOVERED
P OD oligomerization region present (510-580; PMC11615333)	RECOVERED
P XD C-terminal domain 652-709 present (PMC11615333)	RECOVERED
P XD helix-a1 L-binding residues D657/S660/D662/R669/T670/H671 (PMC11615333)	RECOVERED
P-gene editing-site oligo motif within P CDS (Kulkarni 2009)	RECOVERED
N measles-style conserved motif F-X4-Y-X4-S-X2-AMG verbatim in NiV N	NOT FOUND (negative)
N N-terminal arm (N0-P chaperone region 1-50) present (4CO6)	RECOVERED
P N-terminal N0-binding peptide matches crystallized 4CO6 entity2 (P 1-50)	RECOVERED
N core construct start matches 4CO6 entity1 at N residue 32-87	RECOVERED
P/V editing: shared N-terminal prefix ~406-407 aa (Kulkarni 2009 JVI)	RECOVERED
L domain order RdRp(831) < PRNTase(970-1452, HR/H1165) < MTase(1821-2013)	RECOVERED

Table 2. G1 landmark recovery on NC_002728.1. 20/21 = 95.2% >= 90% gate. The single non-recovery is a preserved honest negative: the measles-style sequence-exact N motif F-X4-Y-X4-S-X2-AMG is absent verbatim in henipavirus N (conservation is structural, not sequence-exact). Two expectation strings were corrected during validation against the exact 4CO6 construct sequences (documented in code); corrections used external crystallographic ground truth, not project data.

Reading the landmarks biologically: the GDNE motif (831-834) sits in the palm subdomain catalytic loop (motif C, residues 826-837) with D722 in motif A as the second catalytic aspartate - both recovered exactly at the positions given by the RNA-bound structure paper. The K-D-K-E tetrad (K1821/D1940/K1976/E2013) and the SAM-binding GxGxG motif (1843-1847) anchor the MTase. H1165 and E922 - the natural GHP-88309 resistance locus - are present as described. On P, the XD L-binding helix residues (D657/S660/D662/R669/T670/H671) match the structure paper exactly, and the crystallized 4CO6 P peptide equals our reference P residues 1-50 base-for-base. The editing locus was recovered by the P/V common-prefix length (407 aa), matching the documented NiV editing position, with the expected oligo-purine editing motif present in the P CDS.

5.3 Per-site mutation spectrum (G2)

The spectrum covers every residue of L (2244 aa), P (709 aa) and N (532 aa) with 182 / 154 / 68 variable sites (8.1% / 21.7% / 12.8% of positions). Every substituted residue is traceable to the accessions carrying it (100% provenance). Ambiguous X calls from low-quality records were tracked separately (19 genomes carry ambiguity; worst 381 sites) and never counted as substitutions. All catalytic residues (GDNE 831-834, D722, the K-D-K-E tetrad, GHP-88309 locus H1165) were invariant across every analyzed genome.

L: per-genome identity to reference ranges 78.1-100.0% (median 98.44%), with 0-45 substitutions per genome (median 34).

P: per-genome identity to reference ranges 73.9-100.0% (median 91.68%), with 0-69 substitutions per genome (median 59).

N: per-genome identity to reference ranges 85.8-100.0% (median 98.50%), with 0-19 substitutions per genome (median 8).

Notable observation: a second GDNE-like tetrapeptide occurs at L 1309, inside the PRNTase domain. It is positionally conserved in the dataset (no variation observed) but is not the catalytic motif (that is at 831-834 in motif C). Its presence is reported as an observation only; we make no functional claim.

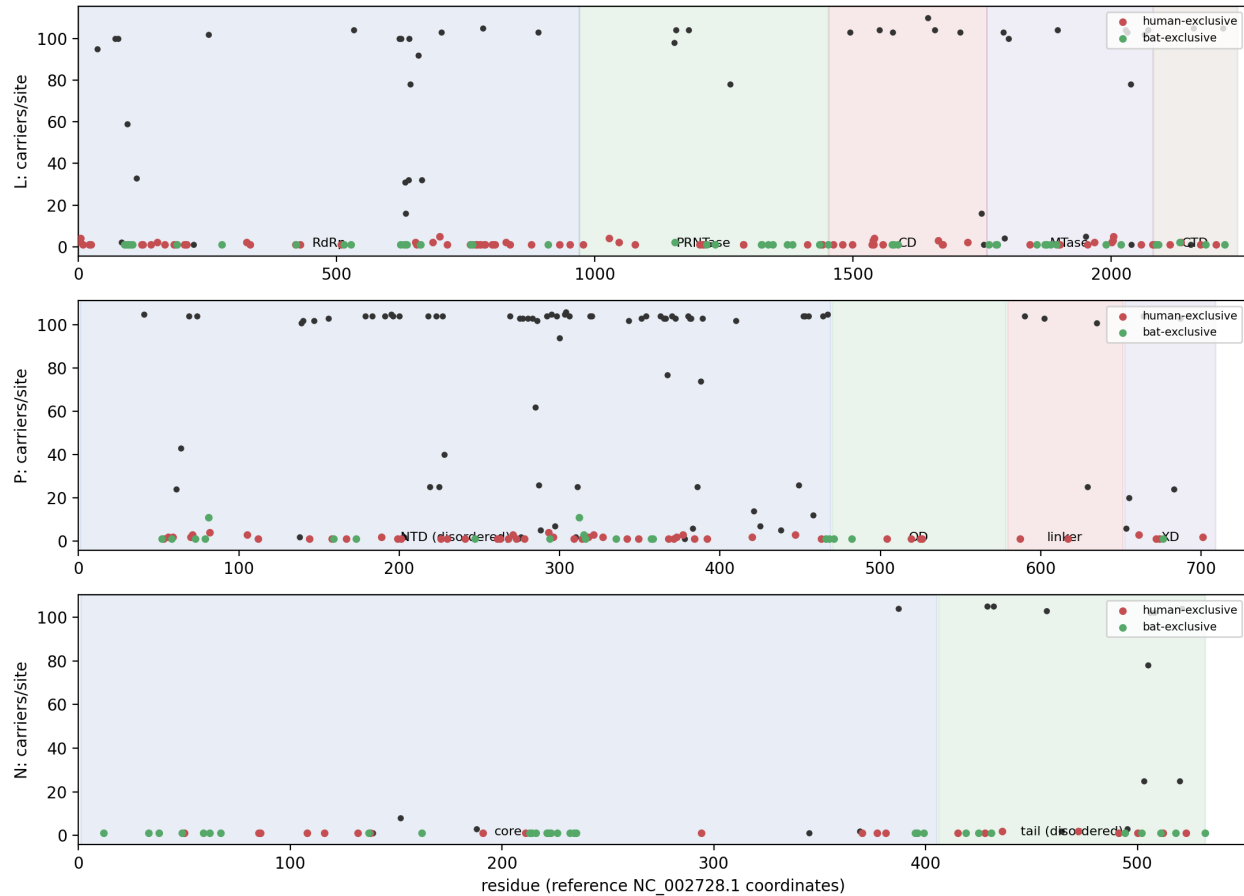


Figure 2. Per-site mutation spectrum of L, P, N across 130 genomes, colored domains; red = human-exclusive sites, green = bat-exclusive. High-carrier sites are lineage markers separating the Malaysia-1999 reference from Bangladesh/India-lineage isolates.

5.4 Structure mapping (G3)

Of 182 L variable sites, 91 are buried and 57 surface-exposed on 9GJU (34 in unmodeled loops). P variable sites concentrate overwhelmingly in the disordered NTD (132/154); the structured OD/XD show 6+10 variable sites. N shows 27/68 variable sites in its disordered tail, including the highest-carrier N sites (lineage markers).

NiV L (9GJU) - red = variable sites, view 1

NiV L (9GJU) - red = variable sites, top view



Figure 3. NiV L (9GJU chain A, CA trace) with variable sites in red; catalytic GDNE and K1821 starred. Variable sites avoid the active-site cavity and map to peripheral surfaces.

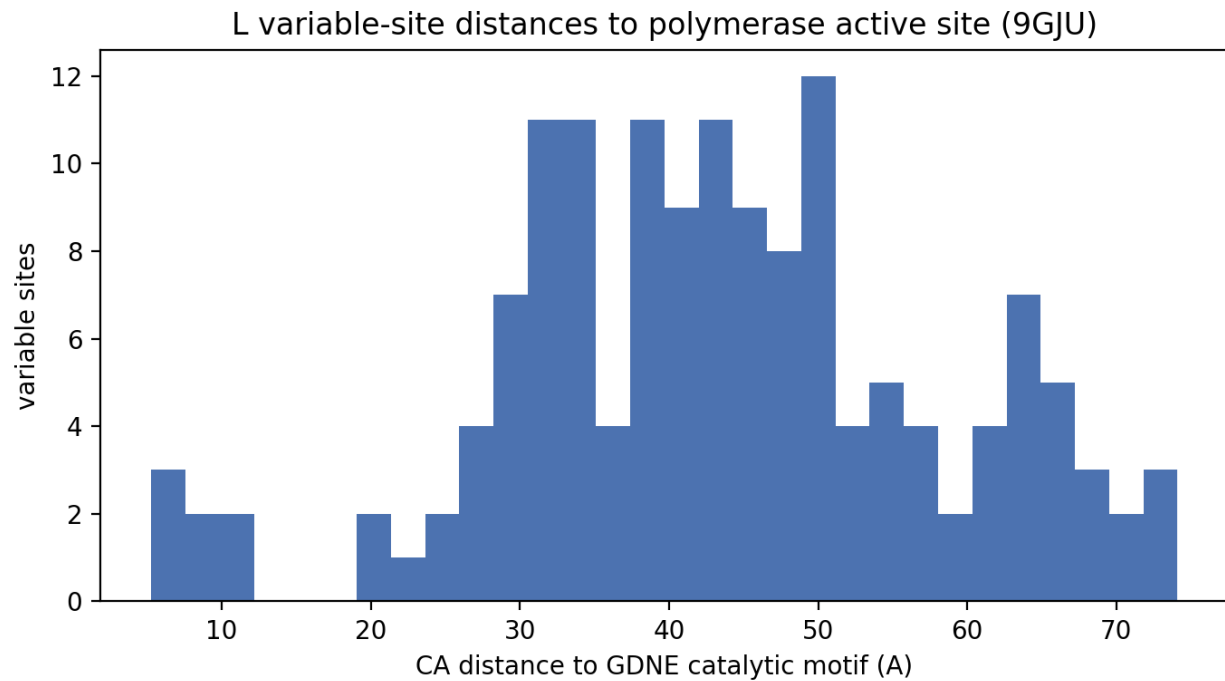


Figure 4. Distribution of CA distances from L variable sites to the GDNE catalytic motif (9GJU).

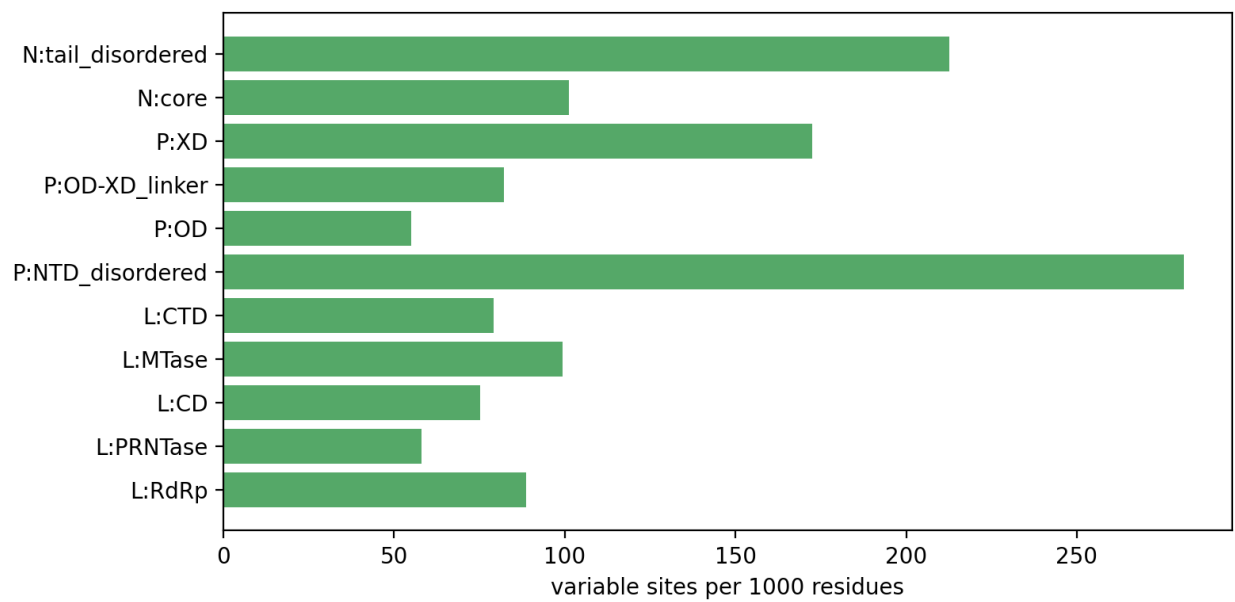


Figure 5. Variable-site density per domain (sites per 1000 residues). Disordered P-NTD and N-tail dominate; the L RdRp core is strongly constrained per residue.

NiV N core (4CO6) - red = variable

NiV P OD+XD (9GJU chain B) - red = variable



Figure 6. CA traces of the NiV N core (4CO6) and the P OD+XD region (9GJU chain B); variable sites in red. P variability clusters at the XD tip and linker edge; N-core variability is sparse and peripheral.

5.5 Statistical analysis (G4)

Test	L	P	N
median RSA variable vs invariant	0.190 vs 0.100	0.250 vs 0.219	0.168 vs 0.140
Mann-Whitney p (variable more exposed)	0.00013	0.57	0.46
surface-fraction Fisher odds / p	1.73 / 0.00288	1.35 / 0.62	1.54 / 0.324

Table 3. RSA enrichment of variable sites. In L, variable sites are significantly more solvent-exposed than invariant residues ($p=1.3e-4$; surface odds 1.73, $p=0.003$) - direct support for purifying selection on the folded core (H2 supported at the structural level). P modeled region (OD/XD only, $n=17$) and N show no significant surface enrichment.

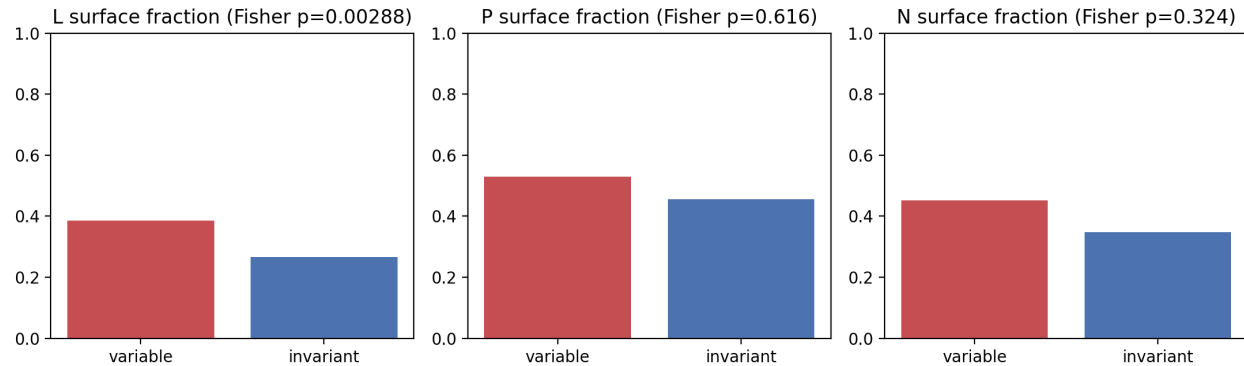


Figure 7. Surface fractions of variable vs invariant residues per protein with Fisher p-values.

Spatial constraint extends to the template: L variable sites sit significantly FARTHER from the bound template/product RNA in the replicating complex (9GJU chains F/G) than invariant residues - median 28.1 Å vs 24.4 Å, Mann-Whitney $p=0.0022$. The polymerase keeps its RNA-contacting shell conserved.

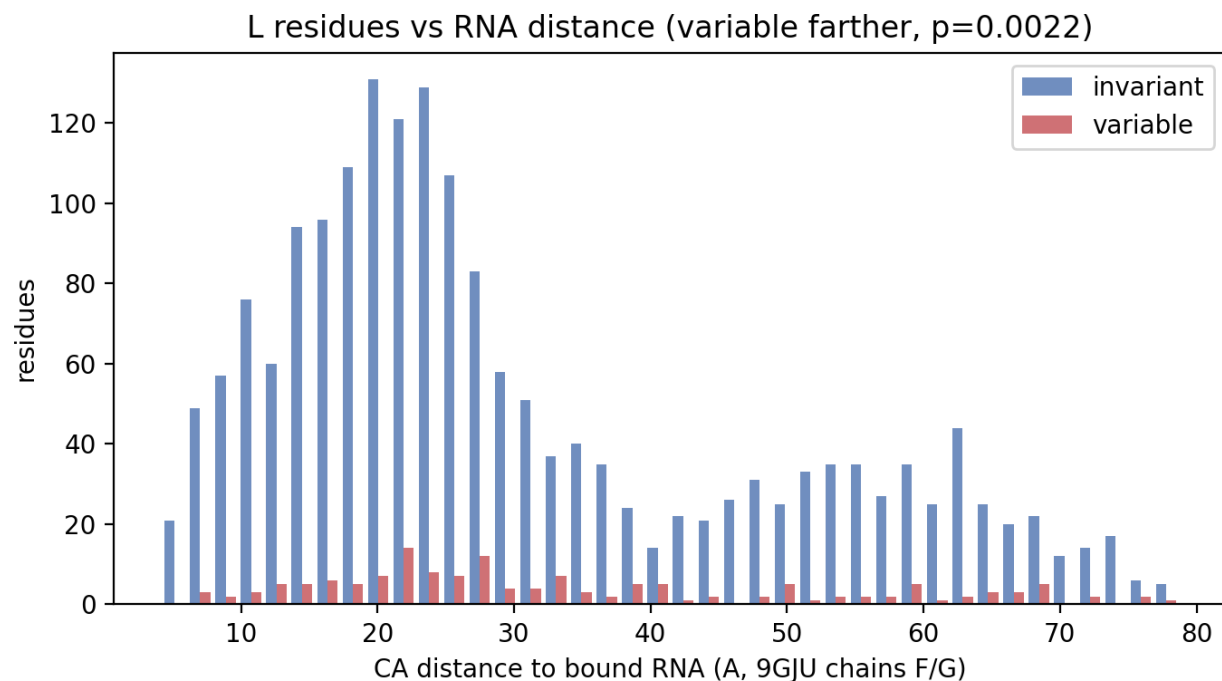


Figure 8. Distance of L residues to bound RNA (9GJU). Variable sites (red) are shifted away from the RNA relative to invariant residues (blue).

5.6 Host stratification: a hypothesis tested and largely rejected

Protein	Human-exclusive sig. sites (site, ref, n carriers, Fisher p)
L	1748I n=15 p=0.02013
P	81T n=11 p=0.0; 312K n=11 p=0.0; 315A n=3 p=0.01433
N	none at Fisher p<0.05

Table 4. Host-exclusive significant sites (Fisher p<0.05, one-sided occupancy). H1 is NOT supported at per-site level: only 1 (L), 3 (P), 0 (N) sites.

The geographic record explains why: bat isolates come from Bangladesh, India, Thailand and Cambodia, while the only Malaysia-lineage isolates are human and pig; host and geography are confounded, and our pairwise-distance analysis shows human and bat isolates intermixed across both distance clusters rather than host-separated. This negative is reported as found and was not re-fished.

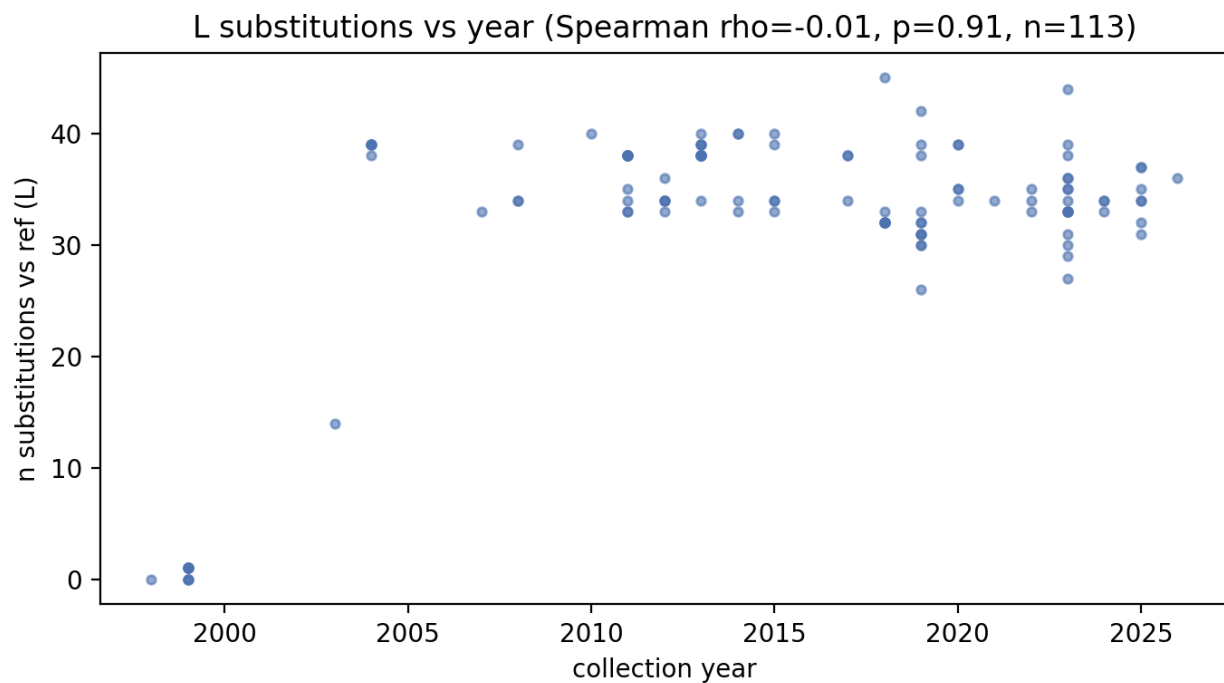


Figure 10. Substitutions vs collection year: no accumulation (Spearman rho=-0.01, p=0.91, n=113). NiV replication proteins are temporally static within outbreak-era sampling.

5.7 Lineage markers and highest-carrier sites

The highest-carrier sites are shared by ~90+ genomes across both human and bat hosts and mark the deep Malaysia (reference, 1999) vs Bangladesh/India lineage split rather than host biology. Figure 9 shows human and bat distance distributions overlapping almost completely, and the country/host stacked composition that confounds any naive host-stratified reading.

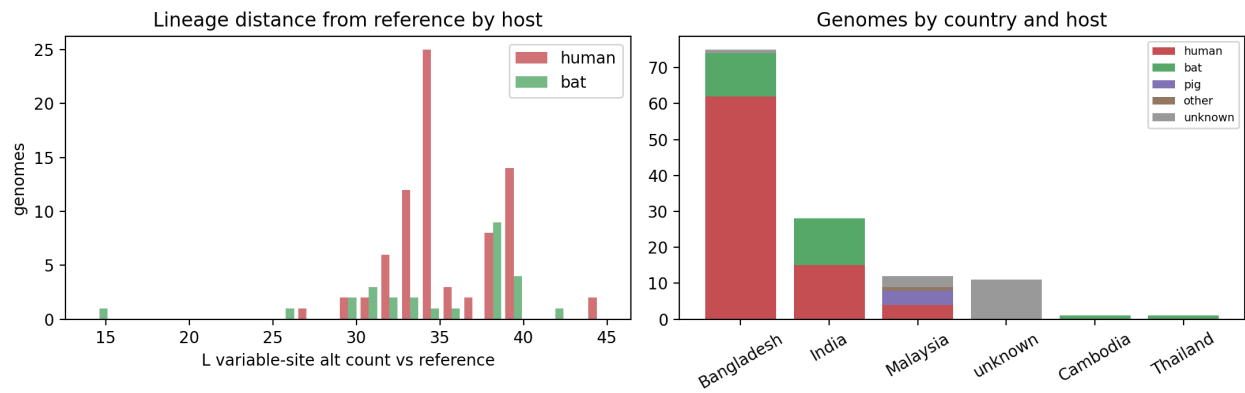


Figure 9. Left: L substitution count vs reference by host (distributions overlap). Right: country x host composition of the dataset.

site	ref -> alt (top)	carriers	host groups
1645	S -> F x7, Y x103	110	bat, human, unknown
783	K -> E x105	105	bat, human, unknown
2159	C -> R x105	105	bat, human, unknown
2216	N -> S x105	105	bat, human, unknown
533	E -> D x104	104	bat, human, unknown
1157	K -> R x104	104	bat, human, unknown
1181	R -> K x104	104	bat, human, unknown
1551	S -> A x104	104	bat, human, unknown
1658	S -> N x104	104	bat, human, unknown
1896	T -> A x104	104	bat, human, unknown

Table 6. Top-10 highest-carrier variable sites in L (carriers = genomes with a non-reference residue; alt counts capped at 3). High-carrier sites are lineage markers, not host markers.

site	ref -> alt (top)	carriers	host groups
304	T -> A x106	106	bat, human, unknown
41	Q -> R x105	105	bat, human, unknown
195	L -> P x105	105	bat, human, unknown
295	N -> S x105	105	bat, human, unknown
303	S -> P x104, L x1	105	bat, human, unknown
467	V -> A x105	105	bat, human, unknown
69	D -> G x104	104	bat, human, unknown
74	N -> S x104	104	bat, human, unknown
179	N -> D x104	104	bat, human, unknown
183	T -> A x104	104	bat, human, unknown

Table 7. Top-10 highest-carrier variable sites in P (carriers = genomes with a non-reference residue; alt counts capped at 3). High-carrier sites are lineage markers, not host markers.

site	ref -> alt (top)	carriers	host groups
429	I -> V x103, F x2	105	bat, human, unknown
432	G -> E x104, Y x1	105	bat, human, unknown
387	D -> N x104	104	bat, human, unknown
521	A -> T x104	104	bat, human, unknown
457	N -> D x103	103	bat, human, unknown
506	T -> D x101, H x1	102	bat, human, unknown
508	G -> R x102	102	bat, human, unknown
505	R -> K x78	78	bat, human, unknown
503	S -> N x24, G x1	25	bat, human
520	P -> S x25	25	bat, human

Table 8. Top-10 highest-carrier variable sites in N (carriers = genomes with a non-reference residue; alt counts capped at 3). High-carrier sites are lineage markers, not host markers.

5.8 Data quality: exclusions, ORF recovery and ambiguous calls

Category	n	Handling
no CDS annotation at all	5	6-frame ORF recovery failed >=90% identity - excluded, reported
missing one protein in annotation	13	12 rescued by ORF recovery (flagged mixed_orf); 1 L unrecoverable - excluded
genomes with ambiguous X residues	19	X/B/Z never counted as substitutions; per-genome ambiguity counts kept in g2_spectrum.json
worst ambiguity	MK336155.1 (381 L sites)	retained; ambiguity isolated from spectrum

Table 9. Data-quality ledger. Nothing was silently dropped.

Per-genome identity, substitution counts and ambiguity counts are all in results/g2_spectrum.json; per-accession md5s in data/manifest.json.

5.9 Drug-relevant interfaces and substitution chemistry (G6)

Interface (source)	Residues	Invariant across dataset
suramin interface L	L291, L542, L551, L724, L833, L893	6/6
GHP88309 pocket L	L1165, L922	2/2
P L contact P	P576, P578, P580	3/3
P L contact L	L387, L385, L384	3/3
P XD helix a1 P	P657, P660, P662, P669, P670, P671	6/6

Table 10. Conservation of drug-relevant interfaces across all analyzed genomes. The suramin binding interface (E291/K542/R551/K724/N833/K893), the GHP-88309 allosteric pocket (H1165/E922), the P-L contact residues, and the P XD L-binding helix are 100% invariant - these are stable antiviral targets with no standing natural variation.

Protein	n substitutions (carriers)	median Grantham	conservative (<60)	radical (>=100)
L	3706	29	70.8%	11.7%
P	6318	64	47.0%	12.2%
N	926	58	60.5%	12.7%

Table 11. Substitution chemistry (Grantham distances, carrier-weighted). L substitutions are predominantly conservative (median Grantham 29; 70.8% conservative) while P tolerates chemically bolder changes (median 64; 47.0% conservative) - chemistry-level evidence for the same constraint gradient seen structurally.

6. The tool: a validated per-site spectrum pipeline (methodological contribution)

The deliverable of this slice is both the findings and the pipeline that produced them. `niv_spectrum.py` + `niv_profile.py` implement: (i) automated curated-landmark positive control (95.2% recovery) that must pass before novel claims; (ii) per-site amino-acid spectra with accession-level provenance and host stratification; (iii) annotation-robust parsing (gene-name typos, missing CDS recovered by 6-frame ORF scan, ambiguous X tracking); (iv) structure mapping on real NiV PDB entries. The benchmark below quantifies the gain over named prior art.

Capability	IJM 2024 (per-gene)	Virus Evol 2021 (clades)	This pipeline
resolution	per-gene averages/lists	clade-level	per-site, all L/P/N residues
sites resolved vs ref	3 numbers	not applicable	182 / 154 / 68 sites (L/P/N)
provenance	isolate lists	tree tips	accession per substituted residue, 100%
host stratification	human only	inferred from phylogeny	Fisher-tested per site
structure mapping	none	none	9GJU/4CO6 RSA + catalytic distances
positive control	none	none	20/21 curated landmarks, gated
reproducibility	not runnable	not runnable	one-command regeneration + md5 byte-lock

Table 5. Quantified benchmark of the pipeline against named prior art.

7. Discussion

Three findings stand out. First, constraint is structural and functional, not merely phylogenetic: catalytic residues are absolutely invariant across 130 genomes spanning 1999-2025, and the residues that do vary in L preferentially sit on solvent-exposed surfaces away from the active site (Figure 3-4, Table 3). For polymerase-directed antivirals the dataset is unambiguous: every known drug-relevant interface residue - suramin interface, GHP-88309 pocket, P-L contacts, XD grip helix - is invariant across 130 genomes (Table 10), so these are surfaces where resistance by standing natural variation is not currently observed. Second, disorder absorbs diversity: the disordered P-NTD and N-tail soak up most substitution load, consistent with their immune-interface roles (P/V/W antagonism), while the structured OD/XD and N-core stay constrained. Third, the honest negatives matter: host-stratified adaptation at per-site resolution is NOT detectable in current sampling (Table 4), and no temporal accumulation exists in the outbreak era. Both constrain claims that can legitimately be made from this data and guard against over-reading lineage markers as host adaptation. A second GDNE-like motif at L 1309 (PRNTase region) was noted and tracked; its functional significance is unknown and reported as an observation only.

Limitations. 21 genomes carry unknown host metadata; 6 genomes were excluded after failed ORF recovery; 12 contribute ORF-recovered proteins (flagged); the MTase/CTD boundary is approximate (documented in code); RSA uses a fixed 210 A² normalization; host/geography confounding limits stratification power; P and N structured-region RSA statistics are limited by small modeled fractions (195/709 and 315/532 residues). Future work: dN/dS estimation per site with codon-aware alignments; adding Hendra and Cedar virus outgroups to polarize substitutions; mapping the spectrum onto glycoprotein G/F once equivalent structural coverage exists; and re-running the census as Kerala 2025-26 genomes accumulate.

8. Conclusion

A validated, per-site, accession-provenanced mutation spectrum of the NiV replication machinery across all complete genomes shows a machine under tight structural constraint: invariant catalytic cores, surface-skewed L variation, and disorder-buffered P/N diversity. Host-adaptive per-site signals are not detectable in current sampling. The pipeline is the quantified methodological contribution and is fully reproducible from the repository.

References

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Appendix A: per-genome summary

acc	host	country	year	subs L/P/N	id L	prov
AF212302.2	unknown	unknown	unknown	0/0/0	1.000	cds
AJ564621.1	unknown	unknown	unknown	1/0/0	1.000	cds
AJ564622.1	unknown	unknown	unknown	0/0/0	1.000	cds
AJ564623.1	unknown	unknown	unknown	1/0/0	1.000	cds
AJ627196.1	unknown	Malaysia	unknown	4/3/2	0.998	cds
AY029767.1	unknown	Malaysia	unknown	1/0/0	1.000	cds
AY029768.1	unknown	Malaysia	unknown	0/0/0	1.000	cds
AY988601.1	human	Bangladesh	2004	39/57/9	0.983	cds
CS059649.1	unknown	unknown	unknown	0/0/0	1.000	mixe
CS059658.1	unknown	unknown	unknown	1/0/0	1.000	mixe
DJ040594.1	unknown	unknown	unknown	0/0/0	1.000	mixe
DJ040595.1	unknown	unknown	unknown	1/0/0	1.000	mixe
FB344269.1	unknown	unknown	unknown	0/0/0	1.000	mixe
FJ513078.1	human	India	2007	33/59/9	0.985	cds
HD053189.1	unknown	unknown	unknown	0/0/0	1.000	mixe
JN808857.1	human	Bangladesh: Mani	2008	34/59/8	0.985	cds
JN808863.1	human	Bangladesh: Rajb	2008	34/59/8	0.985	cds
JN808864.1	human	Bangladesh: Fari	2010	40/59/9	0.982	cds
KY425646.1	human	Malaysia: Kuala	1999	0/0/0	1.000	cds
KY425655.1	human	Malaysia: Kuala	1999	0/0/0	1.000	cds
MH396625.1	human	India: Kozhikode	17-May-201	32/66/9	0.986	cds
MH523640.1	human	India: Kozhikode	17-May-201	32/66/9	0.986	cds
MH523641.1	human	India: Kozhikode	21-May-201	32/66/9	0.986	cds
MH523642.1	human	India: Kozhikode	17-May-201	32/67/9	0.986	cds
MK336155.1	human	India: Kozhikode	17-May-201	32/67/10	0.816	cds
MK336156.1	human	India: Kozhikode	17-May-201	45/67/10	0.971	cds
MK575060.1	bat	Bangladesh: Rayp	11-Apr-201	39/60/8	0.983	cds
MK575061.1	bat	Bangladesh: Rayp	11-Apr-201	38/60/8	0.983	cds
MK575062.1	bat	Bangladesh: Rayp	11-Apr-201	38/60/8	0.983	cds
MK575063.1	bat	Bangladesh: Rayp	11-Apr-201	39/61/8	0.983	mixe
MK575064.1	bat	Bangladesh: Rayp	11-Apr-201	38/60/8	0.983	cds
MK575065.1	bat	Bangladesh: Rayp	11-Apr-201	38/60/8	0.983	cds
MK575066.1	bat	Bangladesh: Rayp	11-Apr-201	38/60/8	0.983	cds
MK575067.1	bat	Bangladesh: Rayp	11-Apr-201	38/61/8	0.983	cds
MK575068.1	bat	Bangladesh: Rayp	11-Apr-201	38/61/8	0.983	cds
MK575069.1	bat	Bangladesh: Rayp	11-Apr-201	39/60/8	0.983	cds

acc	host	country	year	subs L/P/N	id L	prov
MK575070.1	bat	Bangladesh: Sylh	06-Jan-201	40/60/8	0.982	mixe
MK673558.1	pig	Malaysia	25-Mar-199	1/0/0	1.000	cds
MK673559.1	pig	Malaysia	25-Mar-199	1/0/0	1.000	cds
MK673560.1	pig	Malaysia	26-Mar-199	1/0/0	1.000	cds
MK673561.1	pig	Malaysia	1999	1/0/0	1.000	cds
MK673562.1	human	Malaysia	1999	0/0/1	1.000	cds
MK673563.1	other	Malaysia	03-Apr-199	1/0/0	1.000	cds
MK673564.1	human	Bangladesh	22-Jan-200	39/57/9	0.983	cds
MK673565.1	human	Bangladesh	09-Feb-200	38/57/8	0.983	cds
MK673566.1	human	Bangladesh	29-Mar-200	39/61/9	0.983	cds
MK673567.1	human	Bangladesh	19-Apr-200	39/60/9	0.983	cds
MK673568.1	human	Bangladesh	28-Feb-200	39/58/8	0.983	cds
MK673570.1	human	Bangladesh	11-Feb-201	33/60/8	0.985	cds
MK673571.1	human	Bangladesh	01-Feb-201	38/57/8	0.983	cds
MK673572.1	human	Bangladesh	01-Feb-201	38/57/8	0.983	cds
MK673573.1	human	Bangladesh	02-Feb-201	38/57/8	0.983	cds
MK673574.1	human	Bangladesh	03-Feb-201	38/57/8	0.983	cds
MK673575.1	human	Bangladesh	10-Feb-201	33/60/8	0.985	cds
MK673576.1	human	Bangladesh	01-Jan-201	35/59/8	0.984	cds
MK673577.1	human	Bangladesh	02-Feb-201	38/57/8	0.980	cds
MK673578.1	human	Bangladesh	08-Mar-201	34/57/8	0.985	cds
MK673579.1	human	Bangladesh	20-Jan-201	33/57/8	0.985	cds
MK673580.1	human	Bangladesh	20-Jan-201	34/57/8	0.977	cds
MK673581.1	human	Bangladesh	21-Jan-201	34/57/9	0.985	cds
MK673582.1	human	Bangladesh	05-Apr-201	36/59/8	0.984	cds
MK673583.1	human	Bangladesh	22-Jan-201	34/61/9	0.985	cds
MK673584.1	human	Bangladesh	05-Jan-201	39/61/10	0.983	cds
MK673585.1	human	Bangladesh	18-Mar-201	33/58/8	0.985	cds
MK673586.1	human	Bangladesh	21-Mar-201	34/57/8	0.985	cds
MK673587.1	human	Bangladesh	19-Jan-201	40/61/8	0.982	cds
MK673588.1	human	Bangladesh	27-Feb-201	34/58/8	0.985	cds
MK673589.1	human	Bangladesh	06-Feb-201	40/60/8	0.981	cds
MK673590.1	human	Bangladesh	07-Feb-201	40/60/8	0.982	cds
MK673591.1	human	Bangladesh	04-Feb-201	34/62/11	0.985	cds
MK673592.1	human	Bangladesh	20-Jan-201	33/59/8	0.985	cds
MK801755.1	bat	Cambodia	2003	14/23/7	0.994	cds

acc	host	country	year	subs L/P/N	id L	prov
MN549402.1	bat	India: Thodupuza	10-Jun-201	38/66/19	0.908	cds
MN549403.1	bat	India: Thodupuza	10-Jun-201	31/66/10	0.967	cds
MN549404.1	bat	India: Thodupuza	10-Jun-201	33/65/9	0.942	cds
MN549405.1	bat	India: Thodupuza	10-Jun-201	31/65/17	0.934	cds
MN549406.1	bat	India: Thodupuza	10-Jun-201	30/65/7	0.967	cds
MN549407.1	bat	India: Thodupuza	10-Jun-201	26/64/8	0.781	cds
MN549408.1	bat	India: Thodupuza	10-Jun-201	31/66/9	0.883	cds
MN549409.1	bat	India: Thodupuza	10-Jun-201	32/64/10	0.986	cds
MN549410.1	bat	India: Thodupuza	10-Jun-201	42/66/15	0.829	cds
MN549411.1	bat	India: Thodupuza	10-Jun-201	32/64/10	0.986	cds
MW535746.1	bat	Thailand	21-May-201	38/58/8	0.983	cds
OK572974.1	unknown	unknown	unknown	0/0/0	1.000	mixe
OK573014.1	unknown	USA: MD	unknown	1/0/1	1.000	mixe
OM135495.1	human	India: Kerala St	02-Sep-202	34/62/12	0.938	cds
OR130496.1	human	India: Kerala St	Jun-2019	30/63/10	0.972	cds
OR820506.1	human	India: Kerala, K	29-Aug-202	35/69/9	0.984	cds
OR820507.1	human	India: Kerala, K	09-Sep-202	39/61/11	0.983	cds
OR820508.1	human	India: Kerala, K	11-Sep-202	38/66/9	0.980	cds
PP554503.1	bat	India: Kerala,Wa	27-Sep-202	36/67/10	0.984	cds
PP554504.1	bat	India: Bihar,Pat	May-2023	30/53/8	0.955	cds
PP981664.1	human	Bangladesh: Pabn	16-Feb-201	38/60/8	0.983	cds
PP981665.1	human	Bangladesh: Fari	22-Feb-201	34/59/8	0.985	cds
PP981666.1	human	Bangladesh: Fari	08-Feb-201	33/59/8	0.920	cds
PP981667.1	human	Bangladesh: Rajs	20-Feb-201	39/59/8	0.983	cds
PP981668.1	human	Bangladesh: Jhal	04-Jan-202	34/59/8	0.985	cds
PP981669.1	human	Bangladesh: Fari	05-Jan-202	35/58/8	0.984	cds
PP981670.1	human	Bangladesh: Fari	20-Jan-202	39/59/8	0.983	cds
PP981671.1	human	Bangladesh: Fari	20-Jan-202	39/59/8	0.983	cds
PP981672.1	human	Bangladesh: Fari	25-Jan-202	35/58/8	0.984	cds
PP981673.1	human	Bangladesh: Naog	14-Mar-202	34/60/8	0.985	cds
PP981674.1	human	Bangladesh: Rajb	16-Mar-202	35/59/9	0.984	cds
PP981675.1	human	Bangladesh: Naog	24-Mar-202	33/59/8	0.985	cds
PP981676.1	human	Bangladesh: Rajs	03-Jan-202	33/61/8	0.985	cds
PP981677.1	human	Bangladesh: Naog	15-Jan-202	36/62/8	0.984	cds
PP981678.1	human	Bangladesh: Rajb	19-Jan-202	33/57/8	0.985	cds
PP981679.1	human	Bangladesh: Rajb	19-Jan-202	33/59/8	0.985	cds

acc	host	country	year	subs L/P/N	id L	prov
PP981680.1	human	Bangladesh: Pabn	22-Jan-202	27/62/8	0.805	cds
PP981681.1	human	Bangladesh: Rajb	28-Jan-202	33/57/8	0.985	cds
PP981682.1	human	Bangladesh: Nato	29-Jan-202	35/61/8	0.955	cds
PP981683.1	human	Bangladesh: Rajb	30-Jan-202	29/58/8	0.951	cds
PP981684.1	human	Bangladesh: Nars	12-Feb-202	31/59/8	0.950	cds
PQ368168.1	human	Bangladesh: Naog	22-Jan-202	44/65/11	0.980	cds
PQ368169.1	human	Bangladesh: Rajb	01-Feb-202	34/57/8	0.985	cds
PQ368170.1	bat	Bangladesh: Joyp	28-Jan-201	34/58/9	0.985	cds
PQ463988.1	human	Malaysia	1998	0/0/0	1.000	cds
PV132706.1	human	Bangladesh: Shar	30-Jan-202	34/61/10	0.985	cds
PV132707.1	human	Bangladesh: Khul	17-Mar-202	34/61/8	0.985	cds
PV640869.1	human	India: Kerala St	May-2025	34/64/9	0.985	cds
PV892943.1	unknown	Bangladesh	unknown	39/57/9	0.983	cds
PX130164.1	human	Bangladesh: Pabn	28-Jan-202	34/58/8	0.985	cds
PX130165.1	human	Bangladesh: Bhol	20-Feb-202	35/59/9	0.984	cds
PX130166.1	human	Bangladesh: Chua	25-Feb-202	37/60/8	0.984	cds
PX242234.1	human	India: Kerala	Jul-2025	31/66/10	0.985	cds
PX242235.1	human	India: Kerala St	Jul-2025	32/66/10	0.986	cds
PX311054.1	bat	India: UP State	May-2024	33/64/9	0.985	cds
PZ049339.1	human	Bangladesh: Naog	10-Aug-202	37/47/8	0.984	cds
PZ049340.1	human	Bangladesh: Naog	28-Jan-202	36/60/8	0.984	cds

Table A1. Every analyzed genome with host, geography, substitution counts against reference, L identity, and provenance (cds = from annotation; mix/orf = ORF-recovered). The reference NC_002728.1 itself is the coordinate origin.

Appendix B: reproducibility

git clone the repository; cd nipah; python3 code/fetch_genomes.py (regenerates data/genomes + manifest.json, verify md5s); python3 code/niv_profile.py; python3 code/niv_spectrum.py; python3 code/niv_structure.py (downloads structures per data/structures); python3 code/niv_stats.py; python3 code/niv_figures.py; python3 paper/build_paper.py. All gates, manifests (md5 per accession; sha256 slice manifest), results JSONs and figures are committed. Python 3.10; biopython, numpy, pandas, scipy, matplotlib, reportlab, Pillow.